ASK: Explicit, Inspectable Write Decisions for Persistent Agent Memory

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Abstract

Long-term agent memory is usually evaluated by final-answer accuracy, which mixes write-decision quality with extraction, retrieval, answering and judging. We describe ASK, an interface that makes the write decision explicit as one of three actions (KEEP, SUPERSEDE, ABSTAIN) and records each decision with the features, scores, thresholds and identifiers needed to recompute the action. We keep the evidence streams separate. MGP-6 v4.3 is a controlled synthetic decision diagnostic in which the action depends on which two of six pooled sentences the dialogue indexes (verified on the 40 public components); one configuration frozen before hidden scoring reached 0.848 action accuracy on 540 hidden items. An exploratory controller-only LongMemEval-M run on 180 questions scored 128 correct on the grader's output (137 after nine one-sided manual adjudications), and its merged decision log of 12 654 rows is recomputed exactly from the logged features. A separate 2×2 diagnostic contrast on the public development split, run with five substitute configurations of three models, found accuracy differences associated with evidence routing and with the prompt that differ among configurations and, for the Gemini settings, depend on how replies are parsed. We make no comparative downstream or priority claim. The reference protocol was first committed after both evaluations and is not used by any evaluated system; its baseline validator accepted all 19 tampering probes we built, and we supply a corrected validator as a patch. It is a specification, not conformance evidence.

1 Introduction

Agents that keep a persistent memory across sessions face a write problem before they face a retrieval problem. When a new statement arrives, the memory can accept it without replacing anything, replace a stored item with a newer one, or decline to commit because the evidence does not settle the question. These are different operations with different failure costs. In a memory that overwrites in place, replacing a value on weak evidence destroys information, and in any memory, refusing a well-supported update leaves it stale. Common evaluations of long-term memory report the accuracy of final answers [29], which mixes the quality of the write decision with extraction, retrieval, answer generation and judging; recent work argues that final accuracy also hides retrieval and evidence-use failures [17]. A correct answer does not show that the intermediate memory state was correct, and a wrong answer does not show that the write decision was wrong.

Our starting point is that write decisions should be explicit, inspectable and evaluated separately from downstream question answering. ASK is an interface that does this for one decision. It names

three actions: KEEP leaves the compared prior item active and does not replace it (the candidate may also be admitted as an additional record, depending on the store; Section 3), SUPERSEDE replaces a stored item with a successor while retaining the predecessor and a link to it, and ABSTAIN records the decision without changing the stored state. Each decision is written as a record that carries the candidate, the identifiers of the prior and resulting items, the features and thresholds that produced the action, and a rationale string.

We have tried to state only what the evidence supports. The paper makes four bounded contributions, each supported by its own evidence stream.

1. **An interface and a specification.** We describe the decision semantics, the decision rule used in our runs, the decision record, and a reference protocol with a validator and replay-scope declarations (Section 3). The reference protocol was first committed after the LongMemEval run and after the MGP-6 hidden evaluation described below and is imported only by tests. Its validator at the baseline commit accepted all 19 of our self-consistent tampering probes. We supply a corrected validator as a patch that rejects all 19 and a configuration-identity check bound to a frozen configuration; the patch is not part of the baseline commit, was written by us after the evaluations with the help of an AI research assistant, and is not used by any evaluated system. We therefore treat the protocol as a specification and not as evidence of conformance.

2. **A controlled synthetic decision diagnostic with a retained failure.** MGP-6 v4.3 keeps the evidence sentences fixed within a component and varies which sentences the dialogue indexes to the target slot (the numbering of the sentences changes as well; verified on the 40 public components) (Section 4). One configuration frozen before hidden scoring reached 0.848 action accuracy on 540 hidden items. The earlier MGP-6 v2 failed its own construct-validity check and is kept as a record of that failure.

3. **An exploratory downstream run with a decision log.** A controller-only LongMemEval-M run on 180 questions scored 128 correct on the grader's output, the primary count; nine one-sided manual adjudications raise it to 137, which we treat as a sensitivity count (Section 5). The run is exploratory by the terms of its own frozen analysis plan, and we make no comparison with other systems. Its merged log of 12 654 decisions permits exact recomputation of every logged action and score from the logged features (replay level R1); 11 832 of the rows belong to completed question runs and 822 are replayed rows of stopped runs. Higher replay levels are not established.

4. **A diagnostic contrast of evidence view and prompt.** On the public development split, a 2×2 contrast run with five substitute configurations of three models found accuracy differences associated with evidence routing and with the prompt that differ among configurations and, for the Gemini settings, depend on how replies are parsed (Section 4). The manuscript model was not run.

Each element of ASK has prior or concurrent precedent (Section 2); the contribution is the specification, the diagnostic and the evidence record, not the idea of gating, superseding or abstaining on memory writes.

Evidence roles. Table 1 lists the evidence streams and what each can support. We do not combine action-label evidence from the synthetic diagnostic with answer evidence from LongMemEval into one notion of validation, we do not treat exact deterministic replay as correctness, and we do not

Table 1: Evidence streams, what each supports, and what it does not.

Stream	Object	Supports	Does not support
MGP-6 v4.3 (hidden split)	Action labels on controlled synthetic items	Accuracy of one frozen configuration on this construction	Downstream answer quality; human-data generalization
MGP-6 v2	Pilot with a label-determining field	A documented construct-validity failure	Any policy comparison
LongMemEval-M, V12 controller-only run	Answer correctness graded by a model judge, with deterministic guards for 40 answers and a substring fallback for 2	Exploratory accuracy of one controller on a slice reused from tuning	Comparison with baselines; correctness of intermediate memory state
V12 decision log	12 654 logged decisions (11 832 from completed question runs, 822 replayed rows of stopped runs)	Recomputation of the scorecard from logged features (R1)	Replay of state, retrieval or answers (R2 to R7)
MGP-6 v4.3 development split, substitute models	Accuracy by evidence view and prompt for five configurations of three models	A model-dependent split between routing and prompt on the 180 public development items	Any statement about the manuscript model, the hidden split or ASK components

treat a final answer score as evidence about the intermediate memory state. Appendix D lists the status of selected claims.

2 Related work

We group comparators by function and limit each description to what its text, or its released code where we inspected it, establishes. Works marked † were surfaced only by the targeted search and screened against four elements.

Write-time admission and gating. A-MAC admits an extracted candidate when a weighted feature score reaches a threshold, and it describes an update path that removes a conflicting memory and inserts a merged one [34]. SSGM proposes a write validation gate that rejects proposed changes contradicting protected core facts, as a conceptual architecture without an implementation [12]. Persona commits a tentative persona update only when a specification checker finds no hard violation, and otherwise rolls back to the last verified state [14]. Memory-R1 trains a manager over ADD, UPDATE, DELETE and NOOP operations, and a MemGuard framework for memory contamination assigns ADD, UPDATE or SKIP to each memory atom [7, 32]. The CUPMEM prototype in STALE adjudicates whether a stored state is kept, archived, replaced or marked as unknown [3]. Persona (rollback) and A-MAC (rejection) implement no-commit outcomes, whereas the unknown marker in CUPMEM is itself a write, a deferred candidate in TARL enters a pending ledger, and MemTX records quarantined candidates with a reason [3, 14, 15, 30, 34]. Selective Memory†, When Not to Write Memory† and MemReader† also make write-time decisions with archive, needs-review and deferral outcomes [11, 23, 33].

Supersession and belief revision. Graph-native cognitive memory keeps immutable revisions linked by SUPERSEDES edges and relates its revision operator to AGM belief-base postulates [21]. TOKI adjudicates contradictions with typed operators over a bitemporal fact table and writes the losing fact to an audit row [28]. SodaMem applies a deterministic slot rule that closes the older fact's validity interval and writes a SUPERSEDES edge [26]. GPM and the verifier-persisting MemGuard retain replaced records with supersedes links, and ECHO keeps a revision relation over typed bitemporal records [24, 27, 31]. In Governed Shared Memory a detector sets the supersedes link asynchronously after commit [18], and in MemTX the superseded record has no successor-to-predecessor edge in code [15]. Adjacent works concern revocation of stored precedents [35], eviction after unconditional storage [25], or the memory-update gap [22].

Transactional and event-sourced memory and logs. MemTX stages beliefs through a fixed lifecycle with ordered commit checks and machine-checked invariants [15]. TARL separates a learned action predictor from a deterministic executor that acts on accepted, pending and rejected ledgers [30]. Quipu stores signed allow, deny and unknown verdicts with an evidence hash and replays recorded decisions as of a past instant, with a stated limit for denials [2]. The Log is the Agent derives graph state from an append-only event log and distinguishes strict from permissive replay, but it does not govern memory writes [19]. ESAA† replays a hash-verified event log, MemLineage† stores per-write source and context hashes in a Merkle log, and SuperLocalMemory 4.0† writes a hash-chained audit trail atomically with the fact row [1, 6, 20].

Governance, lineage and provenance. MGP specifies a storage-independent and runtime-independent protocol for governed memory writes, with candidate objects, merge hints and an audit sink, and it leaves the write decision to the caller [9]. GPM binds policy and normalizer versions and the ledger head into its decision records for claim release [31], and TOKI pins the decoder tuple that keys its stored judge log [28]. Eywa accepts an extracted fact only after checks against its source text and writes it with a link to immutable evidence [10]. Several close neighbors govern other objects: claim release (GPM), policy compliance of writes (Quipu), persona state (Persona), commit discipline for agent actions (MemTX) and read-time currentness (ECHO) [2, 14, 15, 24, 31].

Evaluation resources and benchmarks. HaluMem separates extraction, update and question-answering evaluation and inspects stored memory directly [4]. MemOps supplies gold lifecycle operation traces but scores them through probe answers rather than against executed write actions or persisted state [8], and MemTrace separates retrieval reach from evidence use [17]. TARL provides human five-way action labels derived from long-memory benchmarks together with exact next-state accuracy [30], and MCB is a synthetic benchmark of the decision to persist, use temporarily, verify or ask for clarification [13]. CPB† scores admission policies, including an explicit abstain action, against fixed gold actions, and MemTxn† reports admission audit, state recovery and downstream F1 separately [5, 16].

Table 2 shows which of six ASK elements the dossiers establish for twelve of the closest comparators, chosen as those with the most feature codes of 2 among the nineteen features that the audit codes at 2 for ASK (a tie for twelfth place at seven codes was resolved toward the earlier-dated work).

The 30 comparators were read in full text with appendices read selectively, so cells marked not stated are provisional, and the 12 search-surfaced candidates were coded on four elements only.

Table 2: Closest neighbors against six ASK elements. Marks come from the audit feature rows: yes, all underlying codes are 2; partial, mixed or lower codes; no, all coded entries are 0; not stated, no code is established. Columns: no-commit abstention; lineage-preserving supersession (replace action, retained predecessor, lineage edge); typed or provenance-carrying records; persistent decision log; decision or transition replay; evaluation of the write decision. Abstention marks include outcomes that write a quarantine, pending or provisional record (MemTX, TARL, MemGuard).

Comparator	Abstain	Lineage	Typed/prov.	Log	Replay	Action eval.
GPM [31]	partial	yes	yes	yes	partial	partial
MemTX [15]	yes	partial	yes	partial	partial	yes
TARL [30]	yes	yes	partial	partial	not stated	yes
Persona [14]	yes	partial	partial	partial	not stated	yes
TOKI [28]	no	partial	partial	yes	partial	partial
Quipu [2]	partial	partial	yes	yes	partial	yes
MemGuard (verifier) [27]	yes	yes	yes	partial	not stated	yes
ECHO [24]	not stated	yes	yes	partial	partial	not stated
STALE/CUPMEM [3]	partial	partial	yes	partial	not stated	partial
SodaMem [26]	partial	yes	yes	partial	not stated	not stated
Eywa [10]	yes	partial	yes	partial	not stated	partial
Graph-native memory [21]	not stated	yes	yes	yes	not stated	not stated

Scope of the comparison. ASK specifies its three-action decision and decision record in a reference protocol that was written after the reported evaluations and is not called by the evaluated systems. Within a documented review of 30 full-text comparators, 12 search-surfaced candidates and 112 targeted search iterations (five of which returned no records because a source failed), whose search record is released, we did not identify a system for persistent LLM-agent memory that implements all of the following: a three-way keep (non-superseding), supersede (lineage-preserving replacement) or abstain (no commit) write decision whose abstention leaves state unchanged; lineage-preserving replacement; per-decision configuration and evidence identity; a declared replay scope; and separate reporting of action-label and downstream evidence. The 12 search-surfaced candidates were coded on four elements, so the conjunction was assessed for them at that coarser level. This is a bounded result relative to the documented search iterations and not a claim that no such system exists. Each element of ASK has prior or concurrent precedent in the works discussed here, and this paper therefore makes no priority claim.

3 The ASK interface

This section describes the decisions, the scorecard rule used in the LongMemEval run, the decision record, and the reference protocol. For each part we state which code produced the reported results and which code is a later specification.

3.1 Decisions

Given a stored state and a proposed write (a candidate), ASK returns one of three decisions.

KEEP The compared prior item, if any, remains active and is not replaced. Whether the candidate is also admitted as an additional active record is a property of the store and not of the decision.

The reference protocol's single-slot mode refuses admission, its multi-record mode admits the candidate only when a flag requests it, and the evaluated LongMemEval runtime admits every non-duplicate candidate on KEEP (11 464 of 11 478 KEEP rows carry a resulting identifier). KEEP therefore denotes a non-superseding decision and not a withheld write, and a count of KEEP decisions is not a count of writes that were withheld. In MGP-6, where no store is materialized, KEEP is a label for the expected preservation of the standing value of the target slot; it tests the decision not to supersede and does not test candidate admission.

SUPERSEDE A specific prior item is replaced by a successor. The predecessor is retained, marked as superseded, and linked to the successor by an identifier and a relation row, within one database transaction in the released code (the memory module used in the evaluated run is not among the files whose hashes the run manifest records).

ABSTAIN No state change is made. The decision is logged with the candidate and the features that led to it, and the record carries no resulting item identifier.

In the MGP-6 diagnostic the three actions are labels on controlled items and no memory state is materialized (Section 4); the two operational meanings of KEEP above are therefore related by the shared property that no supersession occurs, not by identical state transitions.

3.2 Scorecard rule

The controller evaluated on LongMemEval (Section 5) computes five features for a candidate and, when one exists, a selected prior: extraction confidence c, evidence quality e (evidence span specificity and a recency comparison), a marker burden f (0.5 for each negation or hedge marker), a signed relation signal $r \in [-1, 1]$ from a two-stage language-model relation classifier with deterministic routing (negative values indicate that the candidate contradicts the prior), and a type-conditional stability prior ρ (stable preferences above episodic facts above transient preferences). Two scores are computed from these features, S_{rev} with configured weights and S_{abs} without weights, and each memory type has thresholds θ_{sup} and θ_{abs}. The action rule is

$$a = \begin{cases} \text{ABSTAIN} & S_{\text{abs}} > \theta_{\text{abs}} \\ \text{SUPERSEDE} & S_{\text{rev}} > \theta_{\text{sup}} \text{ and } \max(0, -r) > 0 \\ \text{KEEP} & \text{otherwise.} \end{cases}$$

Candidates are produced by a language-model extractor, one turn at a time, and the compared prior is selected among the three nearest same-type memories of other sessions by embedding similarity (sentence-transformers/all-MiniLM-L6-v2) using the relation label. The extractor, relation classifier, answer model and judge are language models; the rule above is deterministic given its inputs.

3.3 Decision record

Each decision is appended to a JSON-lines action log after the database mutation, with a flush to disk for every row in the released code. A row holds a timestamp, the action, the candidate text and type, a candidate identifier, the session identifier, the prior and resulting memory identifiers, the five features, S_{rev}, S_{abs}, both thresholds, and a rationale string. The log is not written atomically with the mutation, so a crash between the two can leave them inconsistent. The record does not contain a configuration digest, an evidence digest or before and after state digests; these fields exist only in the reference protocol below.

3.4 Reference protocol

The reference protocol is a Python specification with an abstract store interface and one in-memory implementation. It defines store modes (single-slot versioned, multi-record, and a temporal-event mode that is an enumeration value without an implementation), a transition function that applies a proposal to a store, a transition event that carries evidence, configuration and before and after digests, a decision trace, an invariant validator, declared replay scopes, and two adapters (for the LongMemEval controller and for the MGP-6 live controller). We tested it as follows and report the result without interpretation beyond what the tests show.

- **Chronology.** The protocol was first committed on 2026-08-03, after the LongMemEval run (2026-07-18 to 2026-07-19) and after the MGP-6 v4.3 evaluation code (2026-07-25). No module outside the test suite imports it, so no reported number was produced through it.
- **Validator.** The five test files that cover the protocol and the memory operations (34 tests) pass at the baseline commit. A full-suite run on a git-archive export of the baseline commit gave 672 passed, 19 skipped and 9 failed; the failures were not investigated one by one, are unrelated to the protocol files in the sampled messages, and are identical with the patch. In an adversarial exercise, the baseline validator accepted all 19 self-consistent tampering probes we constructed, including an altered value or memory type on a retained predecessor, deletion or mutation of an unrelated record, and an unannounced extra record. Separately, the configuration identity is checked for format only: an identity consisting of 64 zeros was accepted when the object was constructed. We then wrote a corrected validator as a patch against the baseline commit (supplement). The same patch was subsequently committed as `4fc32ca` on a separate local audit branch descending from the baseline commit; it is not part of the baseline commit or of the evaluated runtime. It enforces that a transition changes only the records the action allows (the compared predecessor may only be deactivated, the successor equals the candidate, no unrelated record is removed, changed or added), recomputes event identifiers and the trace evidence digest, and verifies a configuration identity against a canonical SHA-256 of the frozen configuration when the caller supplies that configuration (the constructor itself still checks format only). It rejects all 19 probes and the 64-zero identity with named reasons, accepts eight positive controls, leaves the 34 existing tests passing and adds 98 regression tests (132 passing in the five-file subset plus the new file). The accounting requirement that unrelated records stay unchanged is a design choice of the patch, not a condition that the baseline validator checked, and a store that compacts records would violate it. Limits of the patch: the event identifier is an unkeyed digest, so a self-consistent forgery passes; the supplied before-state is checked only against the event digest; the candidate is supplied by the caller; the configuration digest covers the declared mapping and not the code; the temporal mode has no temporal semantics; and neither the evaluated runtime nor any other store has been run through it.
- **Events.** Transition events are returned in memory and are not persisted by the protocol. The evaluated runtime keeps its store in an in-memory SQLite database created for each question, so no store survives a run.
- **Scope.** The temporal-event mode is a placeholder. The adapters are reference code and were not used to produce any reported result.

We therefore describe the protocol as a specification of the intended record and its checks, with a corrected validator that is a regression-tested reference and not a conformance result, and we claim neither conformance of the evaluated systems nor storage or controller independence as demonstrated properties.

4 MGP-6: a controlled synthetic decision diagnostic

MGP-6 is the name of our own memory-governance diagnostic series; it is unrelated to the third-party protocol called MGP in Section 2.

4.1 A retained failure: MGP-6 v2

The first version, MGP-6 v2, contained 180 scenarios, with 90 SUPERSEDE, 66 ABSTAIN and 24 KEEP items. A lookup that is fitted on the training and development splits and reads only the conflict type, without reading the dialogue or the memory state, reproduced the held-out actions exactly. Scenario recognition was therefore sufficient, and no v2 policy comparison can separate reasoning over evidence from recognition of the scenario family. A shortcut baseline that approaches the full policy is a failed construct-validity check, not a competitive result. We keep v2 as a record of the earlier instrument, and every v2 number we cite is diagnostic.

4.2 Construction of v4.3

MGP-6 v4.3 contains 900 items in 100 components in 20 domains, split by component into train (180 items, 20 components), development (180 items, 20 components) and hidden (540 items, 60 components). A component has nine items that share one principal, one memory slot, one standing value and one proposed value, crossed over three dialogue lengths (9, 11 and 13 turns) and three gold actions. Within a component the dialogues of all three lengths are built from the same pool of six evidence sentences, and the three actions differ in which two of the six sentences the dialogue indexes to the target slot; the three indexed pairs partition the pool. The numbering of the six sentences is also permuted between the three actions; the mean indexed position is identical for the three actions within each public split (3.600 in train, 3.575 in development, 3.5875 pooled). We verified the pool structure on the 40 public components (120 length-by-component triples). After the hidden split had already been consumed, we ran a post-hoc structural audit on all 540 hidden construction records. Using proxy components defined by principal, memory slot and standing value, the audit recovered 60 components and 180 component-by-length triples, with no parse failures; all reported Part-A structural checks passed for all 60 components and all 180 triples. The action-balance checks also passed for all 60 components, and the mean indexed position was identical across the three actions. No private component-map file was available, so equivalence between the proxy partition and an official private component map was not tested. This is a post-hoc construction check, not a performance or shortcut analysis. The model-facing record contains an opaque identifier, the stored memory (principal, slot, standing value), the dialogue and an evaluation query, and no field that names a design target or component. Items are described in the dataset card as human-authored and template-guided; we have no independent provenance for authorship.

The construction was checked against surface shortcuts by fitting word TF-IDF classifiers on 12 restricted text views, character TF-IDF classifiers on 2 views, punctuation-only and length-only features, and decision trees of depth 1 to 5 on the training split and evaluating them on the component-held-out hidden split under five seeds. Every probe stayed at or below its preset ceiling (0.40 to 0.65 depending on the view, against a chance level of 0.333); we did not recompute the per-probe accuracies and do not reproduce them here. A cross-split duplicate audit using exact hashes over seven restricted views and a character n-gram similarity threshold of 0.995 passed. Passing a ceiling does not show that no signal remains, and the checks do not show that no shortcut exists.

Two annotation packets covered all 900 items and agreed on all 900 action labels (raw agreement 1.0, kappa 1.0, 300 ambiguity flags raised); their evidence selections never matched exactly (0 of 900

items, because one packet added the first dialogue turn to every selection). Because the labelling rule is deterministic by construction, perfect agreement is consistent with the manual being followable and the items instantiating it; it does not independently validate the construct. The annotation exercise shows that the deterministic action-label manual was applied reproducibly in the two completed packets; it does not establish independent construct validity. The packets were completed by two human annotators independent of the author. The release metadata records the annotator identifiers `ALI` and `MOHAMMAD`; according to the study owner, `ALI` identifies a different individual from the author. A third independent human was designated as adjudicator according to owner-provided study metadata, but no action-label disagreement arose and the recorded label-adjudication count is zero. The builder validates records, hashes and a release attestation, but it does not authenticate reviewer identity or annotation duration.

4.3 Evaluation protocol

The system we call the live controller gives a language model the two messages that the public index in the dialogue points to, together with the stored memory, the requested value extracted from the dialogue, the query, and a prompt that states decision criteria for each action, and then applies a conservative guard that only changes model ABSTAIN outputs when a public-evidence rule resolves them. The model was `qwen/qwen3.5-9b` through OpenRouter, with temperature 0, no reasoning, a limit of 128 new tokens and seed 4301. The configuration, including the prompt hash, was selected on the public development split and recorded in a hashed manifest before the hidden split was scored. According to the study owner, one complete hidden prediction generation was authorized for the frozen controls; it was interrupted once after 432 of 540 predictions and resumed under the same frozen controls. The hidden split has been consumed; we know of no rerun and we did not tune against it. A later successor configuration, developed on the public development split after the hidden aggregate was known (development accuracy 0.95 and 0.961 in the tracked analysis), is not part of this paper's results, and no hidden-split result for it is reported. The provider did not report the routed model revision during development, so the exact model revision behind the hosted endpoint is unknown. The hidden inputs given to the model contained no gold labels, and the run authorization required the predictions to be hashed before the private scorer was used. We omit the cost-matrix analysis that accompanies the release because its weights were committed after the hidden scoring.

4.4 Hidden-split result

Table 3 gives the result for the frozen configuration. Accuracy was 0.8481 (458 of 540 items; Wilson interval 0.8154 to 0.8760). Because items within a component are not independent, we also report intervals from a bootstrap over components (60 clusters; 5 000 resamples, seed 4301): accuracy 0.8037 to 0.8889 and macro-F1 0.8021 to 0.8894. Macro-F1 was 0.8479, Matthews correlation 0.7728 and Cohen's kappa 0.7722. No model fallbacks occurred.

Error structure. The 38 errors on gold-ABSTAIN items split into 30 predicted KEEP and 8 predicted SUPERSEDE, a false-commit rate of 0.211 (38 of 180; Wilson interval 0.158 to 0.276), where we keep the repository's term false commit for any non-abstaining prediction on a gold-ABSTAIN item; for a predicted KEEP it does not imply that a candidate was admitted. Fifteen gold-SUPERSEDE items (0.083) were not superseded, and 11 gold-KEEP items (0.061) were overwritten. Of the 374 items for which the controller did not abstain, 316 were correct (0.845), so conditional accuracy after abstention was not higher than overall accuracy (0.848); this is a descriptive statement and not

Table 3: Frozen live controller on the MGP-6 v4.3 hidden split (540 items, 180 per gold action). Rows are gold actions, columns are predicted actions. Precision and recall refer to the predicted or gold class respectively.

Gold \ predicted	KEEP	SUPERSEDE	ABSTAIN	Recall	Precision	F1
KEEP	151	11	18	0.839	0.795	0.816
SUPERSEDE	9	165	6	0.917	0.897	0.907
ABSTAIN	30	8	142	0.789	0.855	0.821

Table 4: Tracked development-split results for `qwen/qwen3.5-9b` (180 items, 60 per gold action), taken from the later complete development rerun recorded in the tracked analysis. The development run frozen before hidden scoring recorded 0.8556 accuracy (macro-F1 0.8566) for the live controller, and several baseline values differ slightly between the two runs. The live controller is the only system that receives the router-extracted evidence view.

System	Accuracy	Macro-F1
Live controller (with guard)	0.844	0.846
Decision zero-shot	0.339	0.179
B1 no memory	0.367	0.312
B2 flat retrieval	0.356	0.244
B3 recency	0.328	0.217
B4 naive overwrite	0.339	0.189
Ablation without abstention	0.333	0.247
Ablation without typing	0.344	0.190
Ablation without lineage	0.350	0.202
Ablation without provenance	0.339	0.189

evidence of calibration. A predicted KEEP on a gold-ABSTAIN item preserves the stored value; because MGP-6 v4.3 materializes no state, it counts as an error only because the gold label is ABSTAIN. The guard changed 36 of the 202 model ABSTAIN outputs (30 to KEEP and 6 to SUPERSEDE).

4.5 Development-split comparison and what it can support

On the development split (180 items) the tracked results for `qwen/qwen3.5-9b` are summarized in Table 4. The live controller scored 0.844 and the zero-shot policy and four structural baselines (no memory, flat retrieval, recency, naive overwrite) scored between 0.328 and 0.367. This contrast should not be read as an effect of the write policy, for two reasons that we verified in the code. First, only the live controller receives the router-extracted evidence pair; the zero-shot policy and the structural baselines receive the full public payload or a view of it. Second, the live controller also receives a different, longer prompt. The comparison therefore confounds evidence routing with the decision prompt. The four component ablations (without abstention, typing, lineage or provenance) each have their own prompt: three receive the full payload and the provenance ablation receives the dialogue without provenance, so none of them removes one component from the live controller alone.

A 2 × 2 diagnostic contrast on five substitute configurations. To see how much of this difference is associated with each of the two factors, we ran four cells on the 180 public development items: evidence view (full public payload or router-extracted pair) crossed with prompt (one-line prompt or the live-controller prompt), using the repository's own builders and temperature 0. We

call the cells A (payload, one-line), D (routed, one-line), E (payload, live) and B (routed, live); B+g is cell B after the repository's guard on model ABSTAIN outputs. The manuscript model was not available to us, so we used five substitute configurations of three models through a gateway (reached directly for four configurations and through a tunnel for Gemini low; the reasoning levels are gateway aliases): Gemini 3.8 Flash at low, medium and high reasoning, GLM 5.3 Flash, and gpt-oss-120b at medium reasoning. Four configurations were run twice; Gemini at high reasoning was run once. Absolute values are therefore not comparable with Table 4, and the substitute runs do not reproduce the tracked values. The design is not a clean factorial: the prompt factor is a bundle (action definitions, resolvability criteria, abstention criteria, an instruction on the scope of the evidence and an output constraint), the system label line that the repository's prompt builder inserts is not identical across the four cells, and cell E is mismatched because the live prompt text describes two indexed messages that the full payload does not isolate. We therefore treat the experiment as a diagnostic contrast and report associated differences, not effects of the factors. Substitute models were chosen after the Gemini low accuracies had been inspected.

Reply parsing. The reply parser differed among the runs, and the difference matters for the Gemini settings. We use three named rules. *baseline-strict* is the repository parser at the baseline commit, which rejects a reply wrapped in a Markdown code fence. *json-fence-tolerant* additionally removes one surrounding JSON fence before parsing. In the Gemini low run, our own runner extracted the text between the first and last brace; on all stored replies this gives the same action as json-fence-tolerant. GLM 5.3 Flash and gpt-oss-120b ran with the repository parser before it was modified (their runs ended before the modification time of the working copy), and their replies parse identically under both repository rules. The Gemini medium and high runs used a working copy of the repository parser that had been modified, uncommitted, to remove a JSON fence; the exact patch and hashes are in Appendix F. The tolerant rule was introduced after fenced replies had been observed in Gemini outputs, so using it as the primary rule is a post-hoc analysis choice. We chose it because the question is which view and prompt are associated with a correct decision, not whether a model follows an output format. All accuracies below are recomputed from the stored raw replies under each rule (no stored reply exceeds the 120 characters kept, so none is truncated); a reply rejected by the parser counts as incorrect. Under baseline-strict, 49 of 1 440, 471 of 1 440 and 212 of 720 replies are rejected for Gemini at low, medium and high reasoning and none for the other two configurations (Appendix F).

Results. Tables 5, 6 and 7 and Figure 1 give the results under json-fence-tolerant. The intervals in Tables 6 and 7 come from a paired bootstrap over the 20 components of the development split (all nine items of a component are resampled together; 10 000 resamples, numpy `default_rng(0)`, components indexed in order of first appearance in the development file; indices drawn with `rng.integers(0,20,(10000,20))`; one set of indices shared by all contrasts; per-item results averaged over repeats first). Item-level intervals, which ignore the clustering, differ in width (narrower for most contrasts, wider for some) and are in the supplement. We found no pattern that holds in all five configurations.

- Gemini at low and medium reasoning: the live prompt was associated with higher accuracy (differences of +0.07 to +0.12; the interval excludes zero in three of four prompt contrasts), and routing with differences between −0.01 and +0.03 whose intervals include zero.
- GLM 5.3 Flash and gpt-oss-120b: routing was associated with large gains under the one-line prompt (+0.29 and +0.32) and smaller positive gains under the live prompt (+0.08 and +0.17); the live prompt was associated with gains of +0.24 to +0.28 on the full payload and +0.06 to

Table 5: Accuracy on the 180 public development items under json-fence-tolerant parsing (mean of two repeats; one repeat for Gemini at high reasoning). A: payload, one-line prompt; D: routed pair, one-line prompt; E: payload, live prompt; B: routed pair, live prompt; B+g: B after the repository's guard. Gemini rows are Gemini 3.8 Flash at the named reasoning setting; gpt-oss-120b is at medium reasoning. Substitute models; not comparable with Table 4.

Model	A	D	E	B	B+g
Gemini low	0.761	0.792	0.881	0.892	0.928
Gemini medium	0.808	0.811	0.889	0.883	0.922
Gemini high (1 repeat)	0.961	0.917	0.867	0.850	0.900
GLM 5.3 Flash	0.503	0.797	0.783	0.861	0.889
gpt-oss-120b	0.339	0.658	0.575	0.742	0.767

+0.08 on routed evidence. Each difference is smaller when the other factor is present.

- Gemini at high reasoning (one repeat): the one-line prompt on the full payload scored 0.961, the highest cell of any configuration, and the live prompt was associated with −0.09 (full payload; interval −0.16 to −0.03) and −0.07 (routed; interval includes zero). Under the live prompt this configuration predicted ABSTAIN for 84 and 81 of 180 items on the full payload and the routed pair, whereas 60 items have that gold action.

The change from cell A to cell B ranges from −0.11 to +0.40 across the five configurations under json-fence-tolerant. Under baseline-strict it is positive for all three Gemini settings (+0.17, +0.40 and +0.31), so the sign reversal for Gemini at high reasoning depends on the parsing rule; under baseline-strict the full-payload one-line cell of that configuration scores 0.478 and not 0.961, because 93 of its 180 replies in that cell are fenced. With the full payload and the one-line prompt, accuracy ranged from 0.34 (gpt-oss-120b, which committed a supersession for 177 of 180 items in the first repeat) to 0.96 under the tolerant rule. A statement that a full-context or one-line policy is at chance, or that the prompt or the router accounts for the development difference, therefore describes particular models and parsing rules and cannot be carried over from the manuscript model. The guard raised accuracy of cell B in all five configurations (by 0.025 to 0.050), but its separate interval includes zero in four of them.

The development split is the split on which the live prompt was selected, so cells with the live prompt may be optimistic for the manuscript model and need not transfer to other models. The gateway may add text of undocumented content to requests (reported prompt-token counts exceeded the length of our prompts by roughly 2 000 tokens where we could compare, and for the reasoning settings the gateway's accounting may include reasoning tokens), reasoning settings differ among configurations, and the hosted model revisions are unreported. We do not extend the contrast to the hidden split or to the manuscript model, and we make no claim about which part of the prompt matters. Appendix E lists the settings and the runs that were not used.

5 LongMemEval-M: an exploratory controller-only run

5.1 Setup

We ran the scorecard controller of Section 3 on a 180-question slice of LongMemEval-M [29], stratified at 30 questions for each of the six question types (knowledge update, multi-session, single-session assistant, single-session preference, single-session user, temporal reasoning). The answer model was `qwen/qwen3.5-9b` with reasoning disabled, the judge was `google/gemma-4-31b-it`, and candidates were extracted turn by turn from the question's sessions into a fresh in-memory store for each

Table 6: Prompt-associated differences in accuracy (live minus one-line prompt) with 95% intervals from a paired bootstrap over the 20 development components. Both: routed live (B) minus payload one-line (A).

Model	Prompt, payload (E−A)	Prompt, routed (B−D)	Both (B−A)
Gemini low	+0.119 (0.05, 0.19)	+0.100 (0.02, 0.18)	+0.131 (0.04, 0.21)
Gemini medium	+0.081 (0.02, 0.14)	+0.072 (−0.01, 0.15)	+0.075 (−0.01, 0.15)
Gemini high (1 repeat)	−0.094 (−0.16, −0.03)	−0.067 (−0.15, 0.02)	−0.111 (−0.19, −0.04)
GLM 5.3 Flash	+0.281 (0.22, 0.35)	+0.064 (0.01, 0.13)	+0.358 (0.28, 0.43)
gpt-oss-120b	+0.236 (0.16, 0.31)	+0.083 (0.01, 0.16)	+0.403 (0.34, 0.47)

Table 7: Routing-associated differences in accuracy (routed pair minus full payload) with 95% intervals from a paired bootstrap over the 20 development components.

Model	Routing, one-line (D−A)	Routing, live (B−E)
Gemini low	+0.031 (−0.03, 0.08)	+0.011 (−0.05, 0.07)
Gemini medium	+0.003 (−0.06, 0.06)	−0.006 (−0.06, 0.05)
Gemini high (1 repeat)	−0.044 (−0.10, 0.01)	−0.017 (−0.08, 0.04)
GLM 5.3 Flash	+0.294 (0.22, 0.37)	+0.078 (0.02, 0.14)
gpt-oss-120b	+0.319 (0.27, 0.37)	+0.167 (0.07, 0.26)

question. The run used eight shards and three invocations recorded in its manifest; 28 worker processes stopped abnormally (25 runtime errors and 3 interruptions), and the manifest records an exit code of 1 and a status that was never changed from "running". According to the run manifest, which records one control set, and the study owner, the execution was interrupted and subsequently resumed under the same frozen controls rather than treated as a new experimental condition; per-invocation control hashes were not recorded. Result coverage was checked after the run: each of the 180 expected question identifiers (the set attested by the manifest) occurs exactly once in the predictions, answers and judgments, from exactly one shard, with no missing, duplicate or unexpected identifier. Operational metadata of the last invocation are incomplete. The completed outputs were merged from the shards (Appendix A describes how the run directory and the code were verified). The answer provider policy allowed fallback among three providers, and the provider that served each call was not reported. The slice had informed the development of the controller: the analysis plan frozen before the run states that the V12 implementation was selected after inspecting partial V11 outcomes on this slice, and that for this reason the tests, intervals and acceptance readout from this run are descriptive and exploratory. The run's acceptance target of 153 of 180 correct answers was not met.

5.2 Results

The grader scored 128 of 180 answers as correct (0.711): 118 by parsed verdicts of the model judge, 8 by deterministic guards applied before the judge (40 of the 180 answers were never sent to the judge, and 32 of those were scored incorrect), and 2 by a substring fallback after unparseable judge output. After nine manual adjudications described below, the count was 137 of 180 (0.761). Table 8 gives both counts by question type. Exact (Clopper–Pearson) 95% intervals are 0.639 to 0.776 for 128 of 180 and 0.692 to 0.821 for 137 of 180. The post-adjudication report records eight correct answers among the nine abstention questions. The results are for one run. A request seed of 0 was sent to the hosted answer model, but whether the provider honors it is unknown, so determinism of the answers is not established.

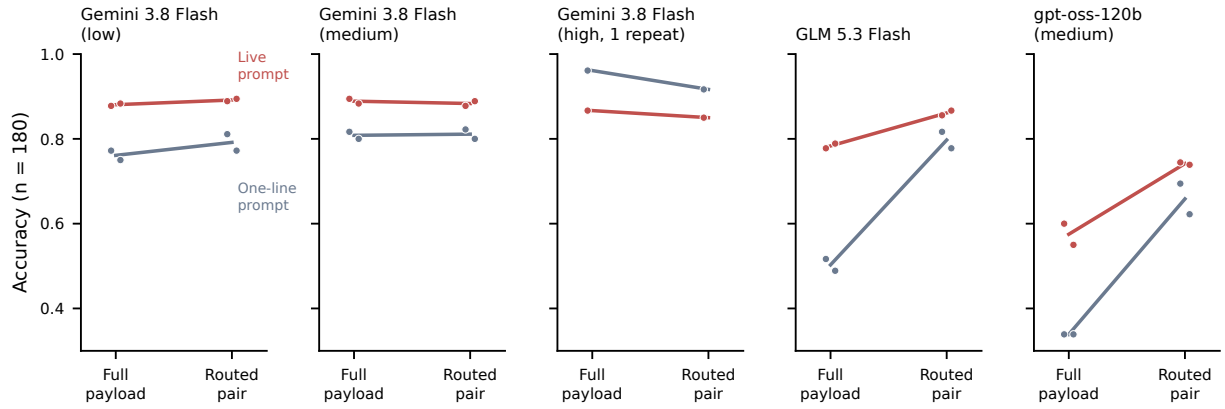


Figure 1: Accuracy on the 180 public development items by evidence view and prompt for five substitute configurations of three models under json-fence-tolerant parsing (repeats shown as points, lines through the means). The sign and size of the routing-associated and prompt-associated differences depend on the configuration; see Appendix F for baseline-strict parsing.

Table 8: Controller-only V12 run on 180 LongMemEval-M questions: correct answers by question type on the grader's output, and after nine manual adjudications (30 questions per type).

Question type	Grader output	After adjudication
Knowledge update	29	29
Multi-session	19	19
Single-session assistant	29	29
Single-session preference	9	16
Single-session user	24	25
Temporal reasoning	18	19
Total (of 180)	128	137

The nine adjudications. Nine judgments were changed by hand from incorrect to correct. All nine concern the controller's outputs, none changed in the other direction, and seven fall in the single-session-preference type (Figure 2). Eight of the nine changed a verdict of the model judge and one changed a verdict from a deterministic guard. The nine identifiers are a hard-coded list in three scripts, each with a short written reason, and the original judge output is kept in backup files. According to the study owner, the nine cases were adjudicated by an independent human reviewer. The available artifacts record neither the reviewer identity nor the review date, and no file records why these nine were chosen or how many of the 52 failures of the controller were reviewed. The same procedure was not applied to the baselines. According to the study owner, their three overrides came from a separate pass by an independent human reviewer. The available artifacts record no reviewer identity, review date or tool, and do not record the selection criteria or review coverage for that pass. The frozen analysis plan asks that results be reported on the raw judge score first and that any adjudication be reported as a separate sensitivity analysis; we therefore treat 128 of 180 as the primary count and 137 of 180 as a one-sided post-hoc sensitivity count, which should not be read as an independently adjudicated accuracy. Seven of the nine reviewed cases were in the preference type; how the cases were selected and how many other failures were examined are not documented, so we draw no conclusion about where the model judge and a human reader disagree.

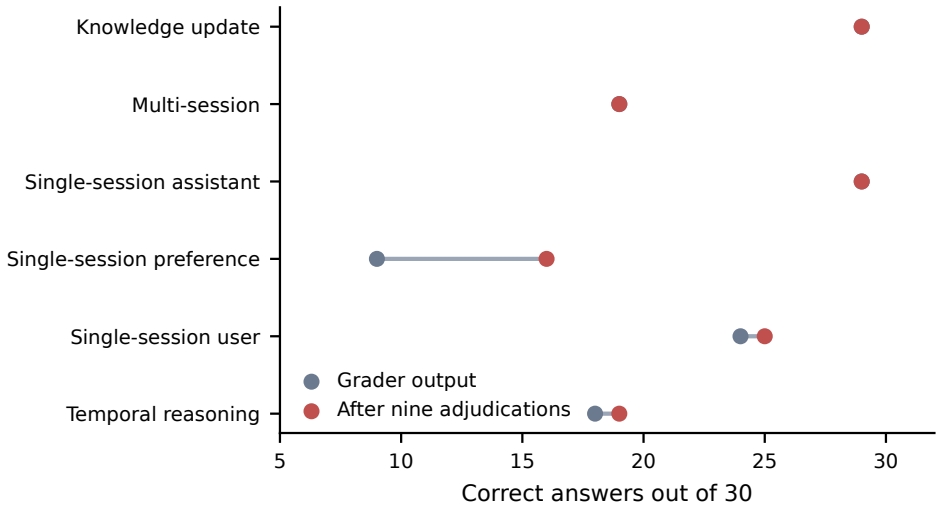


Figure 2: Correct answers out of 30 by question type for the V12 controller-only run, on the grader's output and after the nine manual adjudications.

Why no comparison with other systems is reported. The run directory contains judgments for four baselines (no memory, flat retrieval, recency, naive overwrite). They are byte-identical to the baselines of an earlier run that used a different freeze version, a different shard layout and a different judge configuration (the earlier manifest names a free-tier primary judge with a fallback, whereas its call logs record requests to the identifier without the free-tier suffix, routed to a dated revision), and the model identifiers recorded for the baselines differ in form from those of the controller (a dated slug versus an undated identifier). The V12 run itself evaluated only the controller; its run matrix says that it does not support comparative claims and defines a separate five-system run for that purpose. Three baseline judgments (one for the no-memory baseline, two for naive overwrite) carry override fields. The repository's failure ledger describes comparable manual overrides in an earlier run (v7), made in place and without an executable manifest; it does not document these three. According to the study owner, the three overrides were made by an independent human reviewer. Originals were kept in backup files, but the reviewer identity, date and criteria are not recorded in the available artifacts, and we did not find the tool that wrote them. The plan requires a homogeneity check (one configuration and one judge policy for all systems) before comparative claims. The check as recorded in the run directory returned false because of a stale launcher hash, and it covered the controller only; we found no record of the five-system check required by the plan, and the baseline heterogeneity described above was identified separately. We do not report paired tests between the controller and the baselines. Appendix C lists the baseline counts as context only.

5.3 Decision log and replay

The merged action log of the run holds 12 654 decisions from 3 547 sessions. Of these rows, 11 832 were written by the worker that completed each question and 822 (6.5 percent) are rows of 20 partial question runs that stopped and were replayed after a restart; the log is opened in append mode on resume and the merge step does not deduplicate it. The completed question runs contain 10 712 KEEP (0.905), 373 SUPERSEDE (0.032) and 747 ABSTAIN (0.063) decisions, over 9 192 episodic facts, 2 273 transient preferences and 367 stable preferences; the 822 stale rows add 766, 24 and

32 respectively (all 12 654 rows: 11 478, 397 and 779). All supersessions carry a predecessor and a successor identifier, and no abstention row carries a resulting identifier. Most KEEP rows of the merged log (10 778 of 11 478) have no compared prior. Of the 10 712 KEEP rows of the completed runs, 10 699 carry a resulting memory identifier, that is, the candidate was inserted as a new active record; in this runtime KEEP means a non-superseding acceptance of the candidate and almost never a withheld write (Section 3). Action statistics taken from the merged log include the stale rows unless they are excluded; the exclusion list is in the supplement.

We recomputed the scorecard decision for every one of the 12 654 rows from its logged features, with the scorecard code of the released commit and its default configuration. All actions matched, and the recomputed scores and thresholds matched the logged values with a maximum absolute deviation of 0.0. This establishes replay level R1 (decision-rule replay) for this log, which we bind to the log's SHA-256 hash. R1 shows deterministic recomputation conditional on the logged feature rows; stale rows are themselves outputs of the rule, so R1 holds for them too, and it does not establish that every logged row corresponds one-to-one to an intended extraction event. It also does not establish that the features were extracted correctly, that the store reached the intended state, or that retrieval and answer generation are reproducible; those levels (R2 to R7, Appendix B) are not established. The scorecard modules are not among the files whose hashes the run manifest recorded, so the exact match shows that the released scorecard gives the same decisions and scores on all 12 654 logged feature vectors; we cannot show that the hosted language models would return the same features again.

The extraction trace and cache hold 11 771 validated extracted items, while the merged log holds 883 more decisions than items. The two counts are in different units. In comparable units, 12 654 = 11 832 + 822 and 11 832 = 11 771 + 61. The 822 rows are the stale rows above: each of the 20 stale groups equals, in order, the leading rows of the same question as re-run by the completing worker (822 of 822 rows). The 61 rows are repeated decisions on one cached extraction output that the same shard reuses for two or more questions. The allocation, which rests on stated assumptions about the log and the shard progress files, leaves no row of the 883 unexplained; one extraction-trace row without a cache entry and 16 of 43 repeated cache lookups cannot be characterised at item level, and 23 rows have a cache item whose recorded evidence session differs from the session in which the candidate was ingested.

6 Discussion and limitations

What the evidence supports. The decision record and its R1 replay are verified on one log, whose 883 surplus rows relative to the extraction cache are accounted for (822 replayed rows of stopped runs and 61 repeated decisions on cached extractions). The hidden MGP-6 v4.3 result shows that one frozen configuration classifies the three actions on a controlled synthetic construction with an accuracy near 0.85, the largest share of the errors (38 of 82) being gold-ABSTAIN items on which the controller did not abstain. The LongMemEval run shows that the grader marked 128 of 180 answers correct (mostly by a model judge), and the decision log supports the descriptive statements about decision counts. None of these results compares ASK with an alternative write policy under matched conditions.

Limitations of the specification. The reference protocol was first committed after the LongMemEval run and the MGP-6 hidden evaluation, is not called by any evaluated system, has a baseline validator that accepted all 19 tampering probes and checked the configuration identity for format only, does not persist its events, and defines a temporal mode without an implementation.

The corrected validator is a patch written by us after the evaluations with the help of an AI research assistant; it is regression-tested on probes written by the same party, it has not been run on any evaluated store, and its requirement that unrelated records stay unchanged is a design choice that we have not yet decided to adopt. The evaluated store was an in-memory database per question, and the action log is written after, and not atomically with, the mutation. The released runtime code also differs from the code that produced the LongMemEval run: the run manifest records hashes of 17 code and protocol files, all of which we recovered from repository history and object storage, and the evaluated orchestrator differs from the released one by 912 changed lines after normalizing a package rename. The diffs and a manifest of the 17 files are in the supplement, but the behavioral equivalence of the two has not been shown. Ten of the 17 evaluated files exist in the development repository only as unreferenced objects of a cruft pack that is at risk of removal under default git maintenance (not tested); we recovered them by hash and ship copies.

Limitations of MGP-6. MGP-6 v4.3 is synthetic, template-guided and labelled by a deterministic rule, so agreement between annotators does not validate the construct. The labelling rule makes the action a function of which two pooled sentences the dialogue indexes, and the live controller receives those sentences directly. A high score therefore is consistent with the model, together with the repository's public-evidence guard, applying the decision criteria to the indexed evidence, and it does not measure the ability to find the relevant evidence in free dialogue. We did not show that the distractor sentences cause the errors: the controller on the hidden split never receives them, and a check that every resolvable development error corresponds to a sentence in the same pool that is indexed elsewhere holds by the construction of the pool, so it does not support that explanation. The 2×2 diagnostic contrast used five substitute configurations of three models and the development split on which the live prompt was selected. Its design is not a clean factorial (the prompt factor is a bundle, the system label line differs across cells, and cell E is mismatched), the substitute models were chosen after the first results had been inspected, one configuration has a single repeat, and the primary parsing rule was adopted after outputs had been observed. It found routing-associated and prompt-associated differences that differ among the five configurations and, for Gemini at high reasoning, differ in sign depending on how replies are parsed, so it cannot be transferred to the manuscript model. The substitute-model results do not bear on the hidden accuracy of 0.848, because they come from different models, prompts and parsing rules. The two annotators were independent of the author according to study metadata. The release records use the identifiers `ALI` and `MOHAMMAD`, and the study owner confirms that `ALI` is not the author. A third independent human was designated as adjudicator, but no action-label disagreement arose and the recorded label-adjudication count is zero. These identity and independence statements are study metadata rather than properties authenticated by the release artifacts. Finally, MGP-6 v4.3 materializes no memory state, so its KEEP label describes an expected preservation and not an observed one.

Limitations of the LongMemEval run. The slice was reused from tuning, the run was controller-only, according to the study owner the manual adjudication was performed by an independent human reviewer, although the artifacts record no reviewer identity or date and the case selection is undocumented, and three baseline verdicts carry manual overrides that the study owner attributes to an independent human reviewer, with no reviewer identity, date, tool or selection criteria recorded in the available artifacts, the answer provider was not recorded, a request seed of 0 was sent but provider determinism is not established, and, according to the run manifest (which records one control set) and the study owner, the execution was interrupted and resumed under the same frozen

controls; per-invocation control hashes were not recorded. The manifest records three invocations and incomplete operational metadata for the final one. The judge is a language model, and seven of the nine manual corrections fell in one question type. The final answer score is not evidence about the correctness of the intermediate memory state: a correct answer can follow from a wrong memory and the reverse.

Claims we do not make. We do not claim priority for write governance, abstention, supersession, lineage, event records, replay or layered evaluation (Section 2). We do not claim that the interface is storage independent or controller independent, that it is durable, that it supports temporal queries, that evaluated systems conform to the protocol, that confidence scores are calibrated, or that the write policy causes downstream accuracy. We do not claim a homogeneous comparison of five systems on LongMemEval-M. We also report no hidden-split result for any configuration other than the one frozen before scoring.

Use of AI assistance. AI research assistance (Claude Science, Anthropic) was used broadly in the audit and manuscript-preparation workflow, including assistance with code and evaluation tooling, substitute-model experiments and reanalyses, parser and validator corrections, literature and comparator review, LongMemEval provenance reconciliation, claim checking, figure and table preparation, and drafting. The author reviewed the resulting outputs, reran the reported checks where indicated, and is responsible for the manuscript.

What would strengthen the contribution. A five-system run under one configuration on a fresh slice, with raw results reported first and an adjudication ledger; the evaluated controller run through the reference adapter and validator with persisted events and a corrected validator; and a demonstration that the combination of features supports a capability that the closest neighbors lack, such as conformance checking of a third-party store.

7 Conclusion

We described an interface that names three decisions about a proposed memory write and records each decision with the features, scores, thresholds and identifiers needed to recompute the action, and we reported what the available evidence supports. One configuration frozen before scoring reached an action accuracy of 0.848 on a controlled synthetic diagnostic whose labels depend on which sentences the dialogue indexes (verified on the public components). An exploratory controller-only run on LongMemEval-M scored 128 correct answers on the grader's output (137 after nine one-sided manual adjudications), and its merged 12 654-row decision log is recomputed exactly at the decision-rule level, with 822 of the rows replayed from stopped runs. The reference protocol is a specification first committed after the LongMemEval run and the MGP-6 hidden evaluation, with a baseline validator that accepted non-conforming states, a corrected validator supplied as a patch, and no persistence layer. The main open work is a matched comparison under one configuration on a fresh slice and an evaluated system that runs through the reference protocol.

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A Provenance of the LongMemEval run

The run directory was verified against the output hashes recorded in the repository's derived manifest: the judgments, report and action log of the controller and the judgments and report of the baseline b4 match the recorded SHA-256 values (the derived manifest covers b4 among the baselines). The files of baselines b1 to b3 were verified by byte identity with the corresponding files of the earlier run. The run manifest records 17 code and protocol files by hash. All 17 hashes were recovered from the object store of the development repository and equal the manifest values. Seven blobs are reachable from commits (the only commit holding them at their paths is a tag dated 2026-07-25, after the run). Ten blobs are not referenced by any branch, tag, stash or reflog entry; all ten lie in one cruft pack with a recorded object time of 2026-08-05, and none is present in the public repository clone. Under default git maintenance they are at risk of removal (not tested). The supplement contains the manifest (logical name, SHA-256, object identifier, reachability, relation to the baseline file), the 17 recovered files, the unified diffs against the baseline commit, and a script that archives the ten blobs without rewriting history. We subsequently executed the archive script, and all ten archived blobs matched their recorded hashes. The released orchestrator, longmemeval module, computation module and inference client differ from the recovered versions (the orchestrator by 912 changed lines after normalizing a package rename; the evaluated file has 4 301 lines and the baseline file 4 471). Eleven of the 17 files differ from the baseline-commit file of the same path, two are identical, and four are protocol documents that sit at a different path in the baseline commit. The manifest keys use the package name `ask`, whereas the evaluated launcher builds its list with the earlier package name; we could not explain this difference. We did not test whether the released code reproduces the evaluated behavior.

The result coverage of the controller run was checked as described in Section 5; the merged predictions, answers and judgments each hold the 180 expected identifiers exactly once, and their hashes are recorded in the supplement. The judgments file differs from the concatenation of the shard files in one field of one row (a stored rationale of the earlier pass); verdict fields are identical in all 180 rows.

Table 9: Replay levels and their status for the LongMemEval run.

Level	Name	Status
R1	Decision-rule replay	Established: all 12 654 actions and scores recomputed from logged features and thresholds
R2	Transition replay	Not established: transitions were SQLite statements and no state digests were logged
R3	Serialization replay	Partial, reference code only: events are tested against a schema; no round trip from logged rows to state
R4	Persistent-state replay	Not established: the store was in memory and discarded
R5	Retrieval replay	Not established
R6	Answer-generation replay	Not established: hosted model; a request seed of 0 was sent but provider determinism is not established
R7	End-to-end reproduction	Not established

B Replay levels

C Baseline counts, as context only

Table 10: Correct answers out of 180 for the baselines found in the run directory, as recorded context only. They come from an earlier configuration and are not comparable with the controller; no comparison or test is made, and three of the counts carry manual judgment overrides that the study owner attributes to an independent human reviewer; the available artifacts record neither reviewer identity nor selection criteria.

Baseline	Correct	Source and status
B1 no memory	9	Earlier run (different freeze and judge routing); 10 before one manual override (one item changed from correct to incorrect)
B3 recency	43	Earlier run; no override
B2 flat retrieval	80	Earlier run; no override
B4 naive overwrite	114	Earlier run; 112 before two manual overrides (two items changed from incorrect to correct)

D Claims and their evidence

E Substitute-model runs of the development contrast

Gemini 3.8 Flash at low reasoning was run first, and its accuracies had been inspected before the four later configurations were chosen; those were chosen by availability through one-request tests. Gemini 3.8 Flash at high reasoning was still running when we first analyzed the output directory and is included after completion, with one repeat. None of the completed runs was dropped. Transport failures (two HTTP 503 responses in GLM 5.3 Flash and two SSL errors in Gemini low) were retried and are not counted. The manuscript model `qwen/qwen3.5-9b` was not run.

Candidates that were not used. A Qwen-family hosted model (qwen3.5-plus) returned HTTP 400 on both smoke requests (the error body was not stored; the two records are in the supplement). DeepSeek v4.1 Flash produced 8 records for 5 distinct items. At a first cap of 1 024 completion

Table 11: Selected claims and the status of their evidence in this paper.

Claim	Status	Basis
Frozen configuration reaches 0.848 accuracy on 540 hidden items	Supported as an observation on this construction	Public aggregate; Table 3
Distractor capture explains the errors	Not supported	Distractors are not seen by the controller; check holds by construction
Live controller outperforms baselines on the development split because of the write policy	Not supported	Evidence routing and prompt are confounded; Tables 5 to 7 (substitute models) show configuration-dependent differences
Controller answered 128 of 180 (primary grader result; 137 after one-sided adjudication)	Supported as an exploratory observation	Raw files, hash-bound; Table 8
Controller outperforms naive overwrite on LongMemEval-M	Not claimed	Baselines from a different configuration
All 12 654 decisions replay exactly	Supported at level R1 only	Recomputed from the log, hash-bound
The validator rejects non-conforming states	Not supported	19 of 19 tampering probes accepted
Write governance, abstention or supersession are new	Not claimed	Each element has prior or concurrent precedent

Table 12: Runs used in the 2 × 2 diagnostic contrast. All used the public development split (180 items), temperature 0 and a gateway (local for four rows, through a tunnel for Gemini low). The three Gemini rows are three reasoning settings of one model.

Model and setting	Repeats	Max new tokens	Seed sent	Run window (UTC, 2026)
Gemini 3.8 Flash, low	2	512	no	1 October, 16:59 to 17:16
Gemini 3.8 Flash, medium	2	4096	yes (4301)	2 October, 11:01 to 11:44
Gemini 3.8 Flash, high	1	4096	yes (4301)	2 October, 11:50 to 12:39
GLM 5.3 Flash	2	4096	no	2 October, 09:58 to 10:47
gpt-oss-120b, medium	2	4096	yes (4301)	2 October, 09:37 to 09:55

tokens (inferred from the usage counters, because the cap was not stored in the record) 4 items were attempted: 3 returned no answer after 3 attempts each, with all 1 024 tokens spent on hidden reasoning, and 1 was answered with 725 tokens. At a cap of 4 096 tokens 4 items were attempted, namely the 3 unanswered items and 1 further item, and all 4 returned no answer, with all 4 096 tokens spent on hidden reasoning (finish reason length). Kimi K2.5 was reported to be unavailable and MiMo v2.6 Flash was reported to take 33.8 seconds for a one-line request and was not run; both reports come from our own terminal sessions and there is no raw record for either.

F Reply parsing in the substitute runs

Three parsing rules are involved (Section 4). *baseline-strict* is `parse_action_response` at the baseline commit (file SHA-256 `f1d70129...`), which rejects a reply that is wrapped in a Markdown code fence. *json-fence-tolerant* is the same function with one change: an exact surrounding JSON fence is removed before parsing, and nested fences, surrounding prose and non-object JSON are

still rejected. The change was made in an uncommitted working copy at 2026-10-02 11:00:26 UTC (modified file SHA-256 `9a39ce09...`; a backup file with the baseline hash `f1d70129...` was written three seconds earlier); the unified diff against the baseline commit (44 lines, SHA-256 `6d5deda5...`) is in the supplement as `parser_fence_normalization.patch`. The Gemini low run used the runner's own extraction of the text between the first and last brace; on every stored reply it gives the action of json-fence-tolerant.

Table 13: Parser in use for each substitute run.

Run	Parser at run time
Gemini 3.8 Flash, low (1 October)	Runner's own brace extraction (first to last brace, parsed as JSON); equal to json-fence-tolerant on all stored replies
gpt-oss-120b, GLM 5.3 Flash (2 October, ending 09:55 and 10:47 UTC)	Repository parser before the working-copy change (the working copy equalled the baseline file when the backup was written at 11:00:23 UTC); replies parse identically under both repository rules
Gemini 3.8 Flash, medium and high (2 October, from 11:01 UTC)	Repository parser with the uncommitted fence change

All accuracies in Section 4 are recomputed from the stored replies under each rule (Table 14); no stored reply reaches the 120 characters kept, so none is truncated. A reply that a rule rejects counts as incorrect, and the repository's guard is applied only to valid ABSTAIN outputs.

Table 14: Accuracy of cells A, D, E, B and B+g (defined in Table 5) under json-fence-tolerant (tolerant) and baseline-strict (strict) parsing, and the number of replies that baseline-strict rejects (of all replies for the configuration).

Model	Rule	A	D	E	B	B+g	Rejected
Gemini low	tolerant	0.761	0.792	0.881	0.892	0.928	0
	strict	0.714	0.772	0.856	0.883	0.919	49 of 1440
Gemini medium	tolerant	0.808	0.811	0.889	0.883	0.922	0
	strict	0.406	0.600	0.578	0.808	0.842	471 of 1440
Gemini high (1 repeat)	tolerant	0.961	0.917	0.867	0.850	0.900	0
	strict	0.478	0.767	0.567	0.783	0.822	212 of 720
GLM 5.3 Flash	tolerant	0.503	0.797	0.783	0.861	0.889	0
	strict	0.503	0.797	0.783	0.861	0.889	0 of 1440
gpt-oss-120b	tolerant	0.339	0.658	0.575	0.742	0.767	0
	strict	0.339	0.658	0.575	0.742	0.767	0 of 1440

The tracked results for the manuscript model in Table 4 were produced in July, before the working-copy change. We re-parsed the stored replies of the located set of 12 model-run directories from the earlier July development run (180 replies each): every reply parses under baseline-strict and none is fenced. We did not locate the second set of 12 model-run directories described in the earlier analysis, so this check does not establish parsing invariance for the complete development rerun behind Table 4. We did not inspect the stored replies of the hidden run, which is private; the

tracked analysis records no model fallbacks for it.